The Central Limit Theorem and Inference

GENERAL ASSEMBLY

Agenda

- Some vocabulary
- Random variable review and notation
- The concept of *inference*
- The Central Limit Theorem
- What the CLT isn't
- Hypothesis Testing
- Confidence Intervals
- What CIs aren't

 © 2022 General Assembly

Vocab: Sample vs Population

Just going by the words... what do you think the difference is?

Vocab: Sample vs Population

A **population** (of interest) is every single element that you could ever imagine gathering pertaining to your research question.

A **sample** is a collection of **observations** from your population.

If you end up sampling your entire population, that's called a **census**.

 © 2022 General Assembly

Vocab: Statistic vs Parameter

If you gather sample data and make some calculation on it, you're computing a **sample statistic**. For example, if you take the average of a bunch of numbers, you'll have a sample mean.

If you took another sample, you'd probably get a different sample mean. So what's the "truth"?

If you took the mean of every element of a population, you'd have a **population parameter**.

 © 2022 General Assembly

Notation: Statistic vs Parameter

We generally denote population parameters by Greek letters and sample statistics by letters from our own alphabet. For example:

- Population means are denoted μ (mu) and sample means are denoted with a bar above a letter: $\bar{x}$
- Population standard deviations are σ (sigma) and sample standard deviations with s
- Population proportions with p and sample proportions with $\hat{p}$
- Population regression coefficients with β (beta) and sample coefficients with b

 © 2022 General Assembly

Example 1: Income

What does the average Bostonian make? Suppose you go to Boston, MA to conduct this study. Describe the following:

- What is the population parameter of interest?
- What is the sample statistic you will collect?



Example 2: Medicine

Suppose you are testing a new drug to increase lung capacity. This is measured by having someone take as deep a breath as possible, and having them blow into a machine that measures their lung capacity in liters.

- What is the population parameter of interest?
- What is the sample statistic you will collect?



Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation**
2. **Hypothesis testing**
3. **Confidence intervals**

Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation -** What's the best sample statistic to get as close as possible to my population parameter?
2. **Hypothesis testing**
3. **Confidence intervals**

 © 2022 General Assembly

Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation -** What's the best sample statistic to get as close as possible to my population parameter?
2. **Hypothesis testing -** Is my point estimate statistically significantly different from a proposed population parameter value?
3. **Confidence intervals**

 © 2022 General Assembly

Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation -** What's the best sample statistic to get as close as possible to my population parameter?
2. **Hypothesis testing -** Is my point estimate statistically significantly different from a proposed population parameter value?
3. **Confidence intervals -** If I want to be 95% confident in my answer, what region around my point estimate is appropriate?

 © 2022 General Assembly

Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation -** What's the best sample statistic to get as close as possible to my population parameter?
2. **Hypothesis testing -** Is my point estimate statistically significantly different from a proposed population parameter value?
3. **Confidence intervals -** If I want to be 95% confident in my answer, what region around my point estimate is appropriate?

There is deep theory here, but in general we'll be using the obvious choice. e.g. "sample mean is the best point estimate for the population mean"

 © 2022 General Assembly

Vocab: Inference

Generally, we take samples in order to make **inferences**, or generalizations about a population of interest. There are 3 types of statistical inference:

1. **Point estimation -** What's the best sample statistic to get as close as possible to my population parameter?
2. **Hypothesis testing -** Is my point estimate statistically significantly different from a proposed population parameter value?
3. **Confidence intervals -** If I want to be 95% confident in my answer, what region around my point estimate is appropriate?

We'll talk about these in detail today!

 © 2022 General Assembly

Random Variable Review

Random Variable Review

In your own words, what is a random variable?



Random Variable Review

A **random variable** can be thought of as a *process* for generating values based on some probability distribution.

When real data are collected, you can think of them as **realizations** of that random variable.

Example 1

Tim's Bookstore sees an average of 10 customers enter the shop every hour. What distribution might you assume for this distribution?



 © 2022 General Assembly

Example 1

Tim's Bookstore sees an average of 10 customers enter the shop every hour. What distribution might you assume for this distribution?

$$X \sim \text{Poisson}(10)$$

Example 1

Tim's Bookstore sees an average of 10 customers enter the shop every hour. What distribution might you assume for this distribution?

$$X \sim \text{Poisson}(10)$$

Random variables typically get uppercase letters from the end of the alphabet

 © 2022 General Assembly

Example 1

Tim's Bookstore sees an average of 10 customers enter the shop every hour. What distribution might you assume for this distribution?

$$X \sim \text{Poisson}(10)$$

Random variables typically get uppercase letters from the end of the alphabet

We use this symbol to mean "is distributed as"

 © 2022 General Assembly

Example 1

Tim's Bookstore sees an average of 10 customers enter the shop every hour. What distribution might you assume for this distribution?

$$X \sim \text{Poisson}(10)$$

Random variables typically get uppercase letters from the end of the alphabet

We use this symbol to mean "is distributed as"

A typical way to express the Poisson distribution with parameter 10 (generally different from mean 10, but in this case the same)

 © 2022 General Assembly

Example 2a

In DC during rush hour, the Metro's red line arrives every 7 minutes. What distribution might you assume for this waiting time?



Example 2a

In DC during rush hour, the Metro's red line arrives every 7 minutes. What distribution might you assume for this waiting time?

$$X \sim \text{Exp}(7)$$

 © 2022 General Assembly

Example 2b

In DC during rush hour, the Metro's red line arrives every 7 minutes. How might you express the distributions of this waiting time over *n* days?

Example 2b

In DC during rush hour, the Metro's red line arrives every 7 minutes. How might you express the distributions of this waiting time over *n* days?

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(7)$$

 © 2022 General Assembly

Example 2b

In DC during rush hour, the Metro's red line arrives every 7 minutes. How might you express the distributions of this waiting time over *n* days?

$$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exp}(7)$$

iid = "independent and identically distributed"

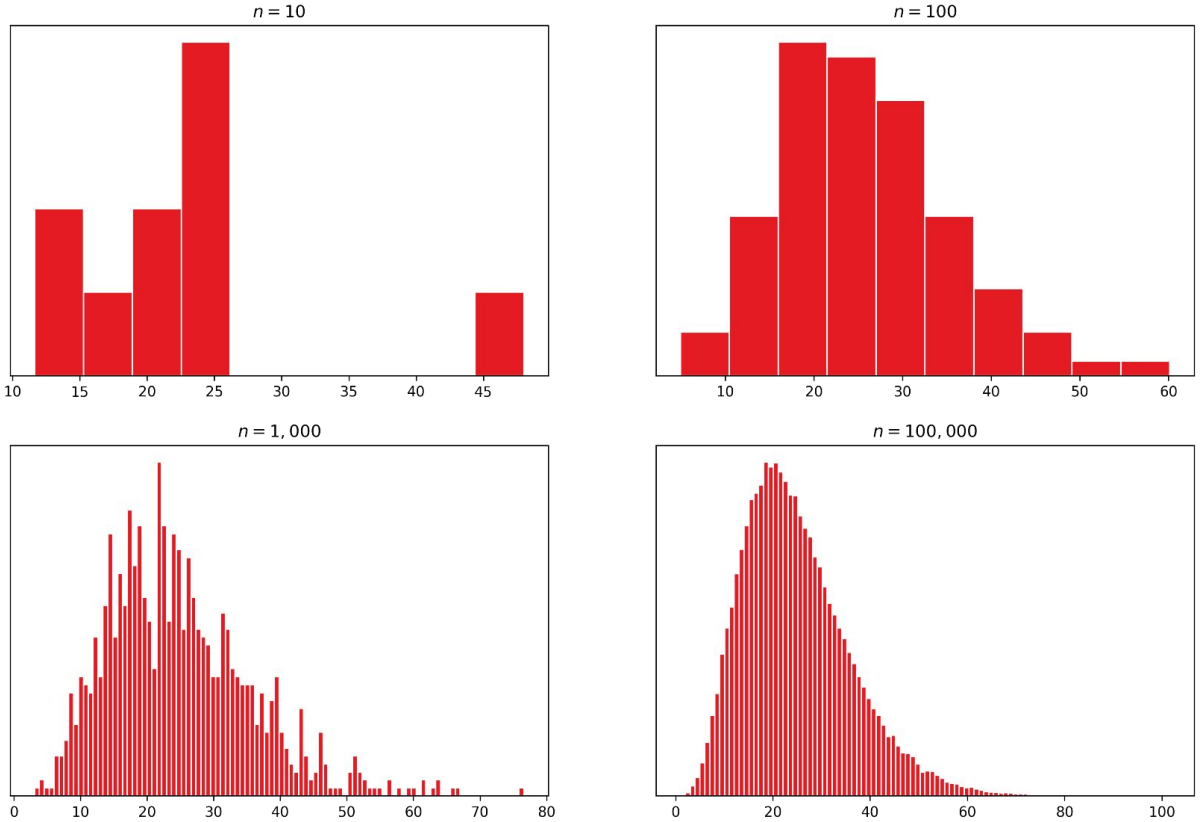
 © 2022 General Assembly

Intuitive Fact: More data, closer we get to the truth

Suppose that X_i belong iid to some distribution. The more data you collect, the more your data $x_1,\ldots, x_n$ will look like the true distribution. That is, the more data we collect, the better picture of reality we get. This makes sense intuitively!

 © 2022 General Assembly

Depiction of Intuitive Fact



 © 2022 General Assembly

Unintuitive Fact: The Central Limit Theorem

From X to X-bar

For a bunch of iid X_is, define the **mean** as:

$$\bar{X} = \frac{1}{n} \sum X_i$$

This will be our **point estimate** of the mean.

The Central Limit Theorem

The mother of all statistics theorems...

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1)$$

where

- μ = population mean of X_i
- σ = population standard deviation of X_i

 © 2022 General Assembly

CLT in Words

The more data we collect, the closer the **distribution of the mean of our data** get to a standard normal distribution.

 © 2022 General Assembly

CLT in Words

The more data we collect, the closer the **distribution of the mean of our data** get to a standard normal distribution.

This is ***MINDBLOWING!*** But why?

CLT in Words

The more data we collect, the closer the **distribution of the mean of our data** get to a standard normal distribution.

We want this large-sample normality of the mean for inference purposes. The normal distribution is well-known and we can do a lot with it!

But first...

CLT Misconception

WRONG:

“The more data we collect, the more our data become normally distributed.”

RIGHT:

“If we collect a lot of data and take the mean, and then repeat this, collecting the mean each time, the distribution of those means is normally distributed.”

Demonstration



 © 2022 General Assembly

Hypothesis Testing

First, a thought experiment:

You want to know if, on average, people are speeding on a highway with a 65mph speed limit. So, you set up a speed camera and record cars' speeds as they pass by.

Using this setup, how can you test whether or not the average car is speeding?



Example: Speeding Cars

To rephrase, your question of interest is:

On average, are cars going faster than 65mph on this highway?

Example: Speeding Cars

To rephrase, your question of interest is:

On average, are cars going faster than 65mph on this highway?

To carry out a hypothesis test, we begin by assuming our question is false and then use data to disprove its falsity.

 © 2022 General Assembly

Example: Speeding Cars

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$

 © 2022 General Assembly

Example: Speeding Cars

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$



The **null hypothesis** is what we assume: the “boring thing” that we wish to disprove with data

 © 2022 General Assembly

Example: Speeding Cars

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$

The **null hypothesis** is what we assume: the “boring thing” that we wish to disprove with data

The **alternative hypothesis** is what we wish to conclude. It’s the opposite* of the null hypothesis

* Ok, you may have noticed that the real opposite of “=” would be “≤” but we don’t do that for two reasons:
1. It’s traditional to write the null with an equal sign
2. It doesn’t work mathematically to assume $\mu \leq 65$ since we don’t have a single number to use in our calculation (this will make sense soon)

Example: Speeding Cars

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$



Why do we use **population parameters** in our hypotheses and not sample statistics?

Example: Speeding Cars

So, we collect data as evidence and get the following:

$$\bar{x} = 72$$



72 is greater than 65. Is this enough information to reject H_0? What are we missing?

Example: Speeding Cars

So, we collect data as evidence and get the following:

$$\bar{x} = 72$$

We need to prove this number is greater than 65 while accounting for statistical variability! That's the whole point of hypothesis testing!

Example: Speeding Cars

So, we collect data as evidence and get the following:

$$\bar{x} = 72$$

So we collected a sample mean of 72. We need to ask ourselves:

- Is this *really* a bigger number than 65?
- ...or was this just by chance due to the random nature of our sample?

The mean is not enough!

Of course, we've collected the data, so we have access to more statistics:

$$\bar{x} = 72$$

$$s = 10$$

$$n = 30$$

 © 2022 General Assembly

The mean is not enough!

Of course, we've collected the data, so we have access to more statistics:

$$\bar{x} = 72$$

$$s = 10$$

$$n = 30$$

BRING IN THE CENTRAL LIMIT THEOREM!!!!!!

The mean is not enough!

Of course, we've collected the data, so we have access to more statistics:

$$\bar{x} = 72$$

$$s = 10$$

$$n = 30$$

The central limit theorem tells us that the sample mean is asymptotically normal. How much data do we need to be "asymptotically" normal? There's no guarantee, but generally the rules of thumb are:

- If n ≥ 30 (arbitrary, I know), or
- If *n* is small, peep the histogram and make a judgment call

 © 2022 General Assembly

But we still need more!

While the Central Limit Theorem holds, there's an issue. Remember, the CLT states:

$$\frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \longrightarrow N(0,1)$$

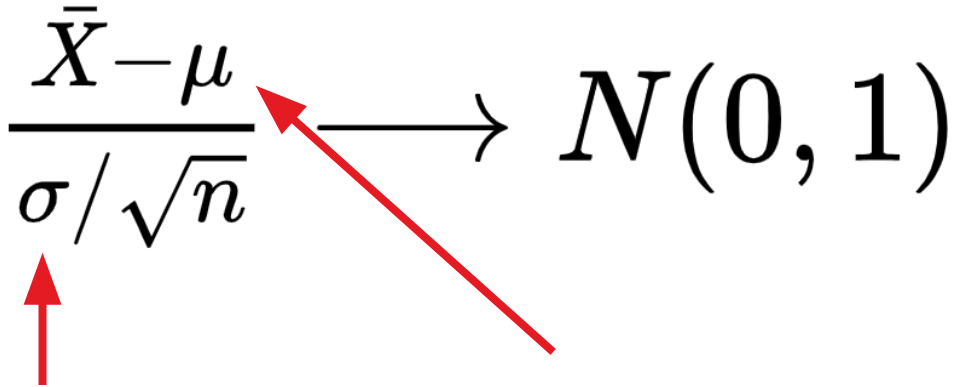


We don't know this! If we want to use *s* instead, we instead need the ***t*-distribution**.

 © 2022 General Assembly

But we still need more!

While the Central Limit Theorem holds, there's an issue. Remember, the CLT states:



 © 2022 General Assembly

Replacements:

So, we replace..

- $\sigma \rightarrow s$, and so we have to use the *t*-distribution
  - Remember the t-distribution is basically the same as the standard normal except a little shorter and a little fatter in the tails.
- $\mu \rightarrow \mu_0 = 65$, since as part of our investigation, **we will assume the null hypothesis is true, and then prove it wrong!**

The one-sample *t*-test of means

And so there we have it! We're doing a hypothesis test of one mean, and will be basing this off of the *t*-distribution.

There are gazillions of types of hypothesis tests out there. We'll stick to a simple one today. The most important one that we'll see in machine learning is one we'll learn about tomorrow, very similar to a one-sample *t*-test.

The Gory Details

To conduct a hypothesis test, this is the general idea:

We assume the null hypothesis is true. We then let our data tell us how likely our outcome was based on the null being true. If the data we got were extremely rare, we conclude that the null hypothesis was false!

The Gory Details



The Gory Details



This is a measurement of “data unlikeliness” based on the underlying distributional assumption. In this case, it’ll be a ***t*-statistic** (based on the *t*-distribution)

The Gory Details



The end result! The p-value tells us how likely our data were to occur under the assumption that the null was true.

***p*-value's low, H_0's gotta go!**

 © 2022 General Assembly

The Results

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$

$$t = \frac{72-65}{\frac{10}{\sqrt{30}}} = 3.83$$

$$p\text{-value} = 0.0003$$

 © 2022 General Assembly

The Results

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$

$$p\text{-value} = 0.0003$$



How did my test turn out?

 © 2022 General Assembly

Result

Classically, there is really only one, very careful, way to announce our results here:

Because the p-value is below 0.05, we reject the null hypothesis and conclude the true mean speed of cars on this highway was greater than 65 mph.

What if...

$$H_0 : \mu = 65$$

$$H_A : \mu > 65$$

$$p\text{-value} = 0.12$$



Interpret my results now...

Result

In this case...

Because the p-value is above 0.05, we fail to reject the null hypothesis and conclude that there is not enough evidence to say that the true mean driving speed is above 65 mph.

A word on *p*-values...

MISCONCEPTION:

"The *p*-value is the probability that the null hypothesis is true."

TRUTH:

"The *p*-value is the probability of seeing the data we saw (or more extreme than the data we saw) under the assumption that the null hypothesis is true."

Um...



SCIENCE EXPLAINERS PSYCHOLOGY

What a nerdy debate about p-values shows about science — and how to fix it

The case for, and against, redefining "statistical significance."

By Brian Resnick | @B_resnick | brian@vox.com | Jul 31, 2017, 12:00pm EDT



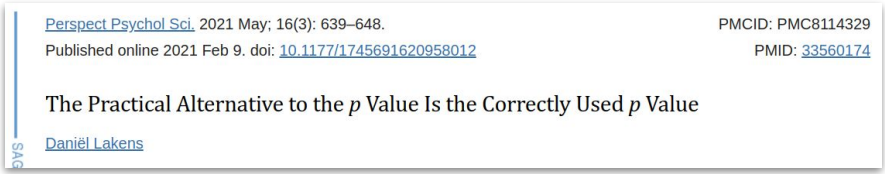
Shanghai Arch Psychiatry. 2015 Dec 25; 27(6): 381–385. PMCID: PMC4858512
doi: 10.11919/j.issn.1002-0829.216027 PMID: 27199532

Language: English | Chinese

Shanghai Arch Ps

The debate about p-values

Ying LU[1,2#*] and Ilana BELITSKAYA-LEVY[1,#*]



Perspect Psychol Sci. 2021 May; 16(3): 639–648. PMCID: PMC8114329
Published online 2021 Feb 9. doi: 10.1177/1745691620958012 PMID: 33560174

The Practical Alternative to the *p* Value Is the Correctly Used *p* Value

Daniël Lakens

SAG



Editorial

Why the *P*-value culture is bad and confidence intervals a better alternative

J. Ranstam

Demonstration



 © 2022 General Assembly

The mother of all hypothesis tests...

ANOVA Tests

(if time... there really should be time)

ANOVA

ANOVA (Analysis of Variance) is sometimes referred to as a hypothesis test, but it's actually an entire category of hypothesis testing.

An ANOVA test tests whether or not two or more means are equal.

The following are the hypotheses of a **One-Way ANOVA**:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \mu_i \neq \mu_j \text{ for some } i \neq j$$

 © 2022 General Assembly

ANOVA

The following are the hypotheses of a **One-Way ANOVA** of three means:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \mu_i \neq \mu_j \text{ for some } i \neq j$$



Why is this not the same as saying $\mu_1 \neq \mu_2 \neq \mu_3$?

 © 2022 General Assembly

Example: Soap

In this experiment, experimenters wish to determine the rate at which three different soaps dissolve in water. Soaps are cut into cubes, weighed, and put into separate portions of a muffin tin and then filled with water. After a specified amount of time, the cubes are removed and weighed.

Soap types:

```
1 = Regular
2 = Deodorant
3 = Moisturizing
```



 © 2022 General Assembly

Example: Soap



To the notebook!

 © 2022 General Assembly

Results: ANOVA Test

We conducted this test for the mean weight loss of 3 different soap types:

$$H_0 : \mu_1 = \mu_2 = \mu_3$$
$$H_A : \mu_i \neq \mu_j \text{ for all } i \neq j$$

And got these results:

$$F = 104.45, p\text{-val} = 5.9 \times 10^{-7}$$

 © 2022 General Assembly

Results: ANOVA Test

$$F = 104.45, p\text{-val} = 5.9 \times 10^{-7}$$

Since our p-value was low, we reject our null hypothesis and conclude that the three soaps exhibit different weight losses when in contact with water.

Which differ, and how? **More on this soon.**

Results: ANOVA Table

The result of an ANOVA test is actually an **ANOVA table**. Being able to read an ANOVA table is a critical statistical skill. The **one-way ANOVA table** looks like this:

	d.f.	Sum Sq	Mean Sq	*F*	*p*-val
Treatment	k - 1	$\sum(\hat{y}_i - \bar{y})^2$	$\frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$	$P(F_{k-1,n-k} > F)$
Error	n - k	$\sum(y_i - \hat{y}_i)^2$	$\frac{SSE}{n-k}$		
Total	n - 1	$\sum(y_i - \bar{y})^2$			

Results: ANOVA Table

The result of an ANOVA test is actually an **ANOVA table**. Being able to read an ANOVA table is a critical statistical skill. The **one-way ANOVA table** looks like this:

	d.f.	Sum Sq	Mean Sq	*F*	*p*-val
Treatment	k - 1	$\sum(\hat{y}_i - \bar{y})^2$	$\frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$	$P(F_{k-1,n-k} > F)$
Error	n - k	$\sum(y_i - \hat{y}_i)^2$	$\frac{SSE}{n-k}$		
Total	n - 1	$\sum(y_i - \bar{y})^2$			

Often omitted in software
(it's the sum of all above rows anyway)

Results: ANOVA Table

The result of an ANOVA test is actually an **ANOVA table**. Being able to read an ANOVA table is a critical statistical skill. The **one-way ANOVA table** looks like this:

	d.f.	Sum Sq	Mean Sq	*F*	*p*-val
Treatment	k - 1	$\sum(\hat{y}_i - \bar{y})^2$	$\frac{SSTR}{k-1}$	$\frac{MSTR}{MSE}$	$P(F_{k-1,n-k} > F)$
Error	n - k	$\sum(y_i - \hat{y}_i)^2$	$\frac{SSE}{n-k}$		

Error between our model and truth

Difference between our model and null model
(ie, high if our model is doing a good job)

Ratio should be high if our treatment was effective!

 © 2022 General Assembly

ANOVA's Dilemma

After looking at graphs, statistics, and doing an ANOVA test, you may now be tempted to say that one mean is higher than another mean - but technically you can't! You didn't test this!

You may then be tempted to perform **individual two-sample *t*-tests**, but you need to be careful! Why?

Think about it...

Suppose that, on any given day, there is a 5% chance of rain. Which is most likely?



A

It will not rain Monday.

B

It will not rain Monday, Tuesday, Wednesday, Thursday, ***and*** Friday.

ANOVA's Dilemma

You're doing the same thing with every hypothesis test you do. You're free to do these tests, but you need to account for these **multiple comparisons** with some **multiple comparisons correction**.

Multiple Comparisons

The simplest (and therefore the most conservative) is the **Bonferroni multiple comparisons correction**. If you're carrying out *m* hypothesis tests, simply replace:

$$\alpha \mapsto \frac{\alpha}{m}$$



Carlo Emilio Bonferroni

 © 2022 General Assembly

Confidence Intervals

Confidence Intervals

A **confidence interval** is an interval around our point estimate that gives us a reasonable bound of what the true population parameter might be.

Confidence Intervals

A **confidence interval** is an interval around our point estimate that gives us a reasonable bound of what the true population parameter might be.

$$\bar{x} \pm \text{something}$$

Confidence Intervals

The interval's bounds are called the **margin of error**.

$$\bar{x} \pm \text{MOE}$$

Confidence Intervals

The margin of error depends on three things:

1. Sample size
2. Sample standard deviation
3. Desired confidence level

Confidence Intervals

The margin of error depends on three things:

1. Sample size
2. Sample standard deviation
3. Desired confidence level



How should each of these affect this interval?
Wider or narrower?

Confidence Intervals

For example, the 95% confidence interval for a **one-sample population mean** is the following:

$$\bar{x} \pm t \frac{s}{\sqrt{n}}$$

Confidence Intervals

For example, the 95% confidence interval for a **one-sample population mean** is the following:

$$\bar{x} \pm t \frac{s}{\sqrt{n}}$$



Can you guess what *t* is? Can you guess its value (usually)?

Confidence Intervals

For example, the 95% confidence interval for a **one-sample population mean** is the following:

$$\bar{x} \pm t \frac{s}{\sqrt{n}}$$



Why s and xbar instead of μ and σ?

Interpretation

The interpretation for a CI is very tricky. Suppose the intervals for one are LB and UB. Here's the interpretation:

 © 2022 General Assembly

Interpretation

The interpretation for a CI is very tricky. Suppose the intervals for one are LB and UB. Here's the interpretation:

I am 95% confident that the true population mean is between LB and UB.

 © 2022 General Assembly

Misconception

WRONG:

“There’s a 95% probability that the true population mean is between LB and UB”

RIGHT:

“I am 95% confident that the true population mean is between LB and UB”

Misconception

WRONG:

“There’s a 95% probability that the true population mean is between LB and UB”

RIGHT:

“I am 95% confident that the true population mean is between LB and UB”



Why?

Demonstration

